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CAD to VR – a methodology for the automated conversion of kinematic CAD models to virtual reality

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Abstract

Driven by the challenges of Industry 4.0 seamless data integration and conversion becomes an even more pressing topic than it already has been in the past. In this context virtual technologies such as Virtual Reality (VR) and Augmented Reality (AR) will play a defining role when it comes down to speed up the development process of new products, factory planning and supporting the workforce with novel intelligent assistance systems to keep up with the faster growing product diversity, especially in the face of the demographic challenge. The key enabling technology to foster this development is the seamless and automated integration of all product data defined in the CAD systems into VR/AR systems. Currently, there is still a major gap in this data integration when more than just plain geometry should be transferred, especially outside of the software ecosphere of a CAD system provider. Considerable effort by VR experts has to be put in to reduce the use case specific complexity of models and to rebuild the already in the CAD system defined kinematic mechanisms and animation. Whilst these obstacles were hindering SMEs in the past to participate of the benefits that virtual technologies provide, this cannot be afforded any longer. In this paper we will outline a CAD and VR system independent workflow for an automated model complexity reduction, animation and kinematic mechanism adoption. We do this by defining light weighted system interfaces and showing solution concepts for each area of the conversion problem. Finally, we verify the presented method on a proof of concept implementation.

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1. Introduction

In response to ever more wide-ranging customer requirements, mechanical engineering products are becoming increasingly complex. In response to this, the software tools installed are also in a state of continuous development. CAD systems are now no longer used exclusively for design but generally also include advanced modules for simulating physical effects (CAE) through to work scheduling (CAP). Furthermore, in the course of the last decade mechanical and motion analyses have become a fixed component of CAD systems. Additional information such as kinematic relationships, which are indispensable for verifying

functionality and collision avoidance, and animations can be created directly in the CAD programs. The utilization of CAD systems by user groups upstream or downstream of design (e.g. service, assembly, marketing, sales) is available only to a limited extent using model viewers, making the assessment of errors/design, for instance, outside the development departments more difficult. The remedy here is the utilization of immersive visualization via Virtual Reality (VR) technology. Important information such as animations or kinematic relationships already present in the CAD program is lost at the moment in the course of the VR data conversion process. These have to be laboriously re-created using special software known as digital content creation (DCC) systems

such as 3dsMax from Autodesk, for instance. This workflow for converting CAD into VR data is extremely complex and is too costly to utilize, particularly for SMEs. The complexity of the design review for functionally designed motion sequences, for example, is very clear to see. In the course of preparing the design review, the user is forced first to export the virtual product data from the CAD system to a suitable CAD exchange format (e.g. STEP) and then to process this with the aid of a DCC program to be finally exported in a VR exchange format such as VRML. During this processing, the kinematic chains and animations lost by CAD exportation must be redefined in the DCC software. For subsequent design changes, this media discontinuity means that changes or updates to the geometry in the CAD system – and possibly to other aspects – are always linked via a manual adjustment to the existing kinematic relationships in VR.

The inhibition threshold for the utilization of VR technology by SMEs can be lowered only by simplifying this workflow by automating the conversion process. In this paper we present a methodology that can solve this issues and has been validated by proof of concept implementation.

2. State of the Art

2.1. Conversion Methods

As Fig. 1 shows, a machine design must currently be processed twice for display in VR. Option 1 relates to a special form which is worth mentioning as it enables stereoscopic video output to a CAD work station, and so represents a rudimentary form of VR technology. Stereoscopic output to a monitor, stands in contrast to the

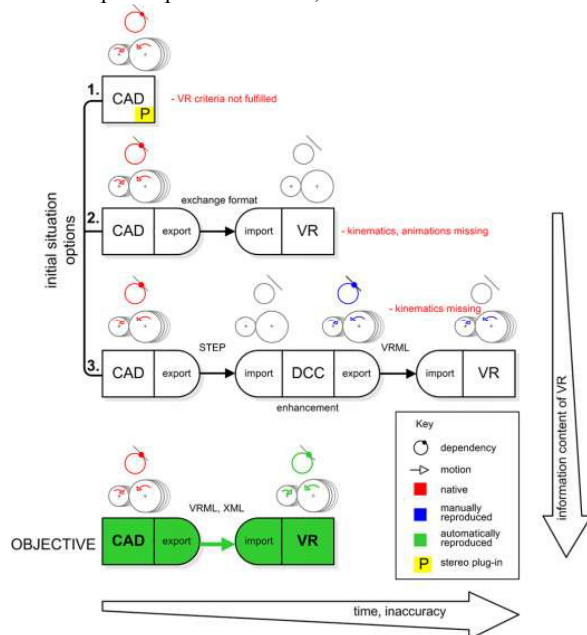


Fig. 1: Transmission of motion data and kinematic relationships to VR

requirements of VR to be a collaborative tool and to display the model on a scale of 1:1. Thus options 2 and 3 are the two currently practicable yet time-consuming routes for creating VR models from CAD data. However, in option 2, the VR model lacks in every case the kinematic relationships and animations such as exploded views (shown in red in Fig. 1) which are completely defined in the proprietary CAD format. In order at least to be able to present defined animations in VR, additional DCC programs must therefore be installed (option 3). The position and orientation of each individual part for each stage of animation must therefore be saved in the VRML file. The non-transfer of further meta-information from the CAX system is not addressed here.

Graf et al. [1] were the first that presented a method for an automated conversion process of CAD data to VR using a product data management system (PDM). They introduce a VR reference model map of the assembly and part structure of the CAD model to reference the VR model. The generation of the VR reference model map as well the VR model is done in an automated way.

In 2009, Stelzer et al. [2] proposed a solution to the conversion problem based on the storage and management of STEP and JT data such as geometry, material and product information in a PDM [3]. With the help of a VR session manager [3] developed in this context, the CAD data saved in this way can then be displayed in VR, which also facilitates interactive downloading of further VR assembly groups directly from the PDM system. Their main objective was the development of a VR-based collaborative platform which would enable the loading of assembly groups from various CAD systems [2].

Raposo et al. [4] and Schilling et al. [5] suggested similar approaches, which also had the objective of converting static CAD data for VR use, with the help of developed software components. The disadvantage of these approaches is that defined animations and kinematics can only be used in the context of the functionality provided by the installed VR software.

Kim and Weissmann [6] present a similar approach as Stelzer et al. [2, 3] for the conversion of CAD models from multiple CAD systems to VR by introducing a middleware approach also integrating a PDM system as the central data storage component.

Tang and Gu [7] introduce an approach for the conversion of CAD models to VR models. Their conversion method reduces the mesh size significantly compared to the standard CAD exporter. During the conversion process the information of design hierarchy consisting of assemblies and parts gets lost. Also no animation or kinematic relations defined in the CAD model gets transformed.

2.2. 3D Data Exchange Formats

The key question when considering 3D data exchange formats is this: Is there a 3D data format which includes geometry, materials, textures, animations, constraints and kinematic relationships and which can be exported by most

CAD systems and imported by every VR system? Table 1 shows that STEP is in theory the answer to this question, but looking at the implemented parts of the STEP standard in CAD system STEP drops out as the answer. When considering the whole STEP standard (ISO 10303[8]) there are various STEP parts, particularly Parts 105 and 108, with which the requirements can be met. However, at the present time only a fraction of the defined STEP parts have been implemented in CAD and VR systems. Other widespread data exchange formats like JT or IGES do not go beyond the transmission of static information.

One exception is VRML 2.0. Although this is a relatively old format and an extended follow-up format X3D has been available for a long time, as Table 1 shows it comes closest to meeting the requirements. VRML represents geometry, materials, textures, animations and to a certain extent interactive kinematics. Furthermore it can be imported by any VR system and exported by a wide range of CAD systems (SolidWorks, PTC Creo Elements, NX, CATIA, Inventor etc.). There are, however, still limitations in terms of the CAD export, as often only geometry, textures and materials are represented.

Table 1: Characteristics of 3D data exchange formats relevant to CAD-VR conversion

	JT	STEP	IGES	VRML
Geometry	Yes	Yes	Yes	Yes
Materials	Yes	Yes	Yes	Yes
Textures	Yes	Not implemented in CAD exporters	No	Yes
Animations	No	Not implemented in CAD exporters	No	Yes
Constraints	No	Not implemented in CAD exporters	No	Limited
Kinematic relationships	No	Not implemented in CAD exporters	No	Limited

2.3. 3D-Model Reduction

3D models which can be rendered in real time are necessary for presenting in VR. It may be inevitable, therefore, to reduce the VR model for large, detailed CAD models. This is usually supported by special software within a semi-automated process. A good, fully automated solution with a balanced ratio of a high level of reduction to the necessary representation of detail and which attains the results of an expert does not exist at the moment.

There are basically two simplification methods. The first aims to reduce the surfaces to be rendered by coarsening the tessellation and removing geometry not relevant to the application. The second simplification method attempts to reduce the complexity of computation by optimizing the 3D structure through bundling similar parts or flattening hierarchies, for example.

Examples of special software for model reduction are NuGraf/PolyTrans from Okino Computer Graphics and GPure from DeltaCAD. As the direct comparison of NuGraf/PolyTrans and GPure in Table 2 shows, the first

stands out above all for its excellent import/export functionality which (particularly in VRML format) includes all information. The options for model reduction are, however, limited to polygon reduction and hierarchy simplification, whereas functionalities for geometry simplification/optimization are not implemented. The latter is possible using GPure and, in conjunction with polygon reduction and hierarchy simplification, more intensive model reductions can be achieved. Negative aspects of this software are the relatively small scope of supported 3D formats in import/export and, in particular, the loss of cameras, lighting and animations in VRML export.

Table 2: A comparison of the most important characteristics of NuGraf/PolyTrans and GPure

NuGraf/PolyTrans	GPure
Simply polygon reduction and hierarchy simplification	Model simplification, polygon reduction and hierarchy simplification
Supports almost all 3D data formats	Supports fewer 3D data formats
Receives all VRML information including animations	Only VRML geometry exported; animations, cameras and lighting are lost

3. Objective

For the effective and time-saving conversion of CAD models to VR models in the SME environment, five basic targets or functionalities can be deduced:

1. The generation of a '1-click' static VR model including fully automated model reduction.
2. The generation of a static VR model with a few easy-to-operate setting options in relation to the model reduction quality.
3. The automatic construction in the VR model of the animations defined in the CAD system.
4. The construction of interactive kinematics as defined in CAD on the basis of mechanism functionality.
5. The saving to the PDM system of all relevant data generated during the conversion process.

A summary of the objective is illustrated in Fig. 1. It is desirable that no further intermediate or supplementary models should be created in the CAD or VR systems, in order to keep the complexity of synchronization and updating to a minimum.

4. Conversion Concept

4.1. 3D Model Reduction

Automated model reduction is an extremely complex problem. The diversity of possible product geometries (from refrigerator to machine tool) is indicative of the challenge. Due to this exceptionally complex parameter, a partly or fully automated solution is likely to never achieve the quality of a

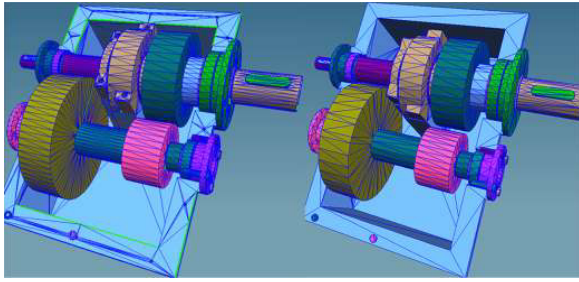


Fig. 2: Example of model reduction using GPure: original model (left) and reduced model (right)

reduction carried out manually by experts. Nevertheless, a solution which is adequate for a number of applications can be found with the help of specialized software.

If this type of specialized software is to be used as a basis, it must meet three principle requirements:

1. Script or Application Programming Interface (API)-based accessibility for automated use
2. Wide-ranging reduction function
3. Numerous supported 3D data formats in the import/export area

The programs already mentioned – NuGraf/PolyTrans and GPure – meet these requirements, as a result of which they have been examined in relation to

- which reduction functions with
- which parameter settings and
- which results

they are able to provide. Fig. 2 makes clear using the example of a working model that, by using automatic model smoothing, closing irrelevant holes and subsequent polygon reduction, it is possible to reduce the number of surfaces by approximately 50 % with minimal optical loss.

4.2. Animation Transition

Because of proprietary characteristics, there is no uniform method for the transfer of defined animations, e. g. exploded views, out of the CAD system; instead, there are three principle solution concepts:

1. Animations are exported by the CAD system exporter into VRML format.
2. The motion information for the objects is captured in a file on a time step basis.
3. The motion information for the objects can be recovered via the CAD API.

Depending on the CAD system under consideration, the path which can be embarked on and how the particular implementation is to be designed must be precisely determined. For example, with the CAD system NX, the first

solution can be taken into consideration whereby the limitation exists that currently only one animation can be exported to the VRML. In contrast, this functionality is currently not implemented in the case of the CAD system PTC Creo Elements. A solution is available here via the second concept, as the transformation matrices of the objects moved are stored in a file on a time step basis. The animation can then be constructed automatically from this information in the VRML.

4.3. Interactive Kinematics

When interactive kinematics are converted to VR the mechanism functionality from the CAD system should be reproduced, i.e. dragging an object should result in the movement of all other linked objects as per the defined kinematic relationships. Having this functionality in VR would be an enormous advantage, particularly for design review as, combined with immersive visualization, collisions could be easily recognized.

In order to implement this, three programming conditions have to be met:

1. It must be possible to ‘grab’ a single object in VR and to capture the motion changes in the context of the six degrees of freedom during run time.
2. The VR must enable a change of position and orientation of the objects during run time.
3. The kinematic relationships between the objects must be accessible from the CAD system.

VRML fulfills the first condition and all professional VR software the second. The fulfillment of the third condition is depends on the specific CAD system but can be assumed.

It is not possible to reproduce the kinematic dependencies of objects as they are defined in CAD completely using VRML. Solutions explored include the installation of a physics engine and of simulation software, the use of a kinematics module of a VR software system, and an appropriate self-developed solution. Although each of these concepts might provide a solution, they must nonetheless be rejected for the following reasons:

1. The addition of a third model in addition to the CAD and VR models
2. Implementation costs are too high

The chosen solution is therefore to use the CAD system itself for the kinematics calculation. The three most important reasons for this are:

1. Comprehensive reproduction of all kinematic relationships.
2. The two existing models (CAD and VR) are sufficient.
3. The implementation costs are reasonable.

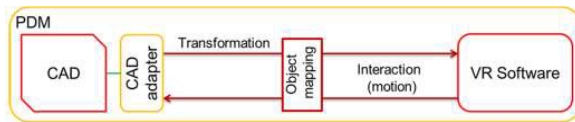


Fig. 3: Solution concept for interactive kinematics functionality

This solution has general validity as a general interface has been defined which must be implemented in a CAD system or in a suitably extended module. This interface must transfer the marked object together with the translations and rotations performed on it by an interaction device as parameters from the VR to the CAD system. The CAD system then returns a result to the VR in the form of the transformation matrices of all interdependently moving objects, i.e. all parts integrated in the kinematic chain receive new position and orientation information.

This interface is implemented with the help of the CAD API. Proof of concept implementation of this principle has been carried out using the Siemens NX and PTC Creo Elements CAD systems as well as using the VR software Instant Reality. Fig. 3 shows the schematic diagram of the solution concept including the CAD adapter which performs the interface functionality. The bi-directional representation of

a CAD model element on a VR model element is carried out via object mapping.

5. Conversion Workflow

The solution concepts for the three problem sets of model reduction, animation and interactive kinematics were introduced in section 4. These must be linked within a workflow in such a way that a VR model can be produced automatically. Fig. 4 shows the conversion workflow diagram. The model is first of all exported from the CAD system and fed through the model reduction process. The animations must then be built up in a converter and be enhanced by other elements which are essential for “grabbing” for the purposes of interactive kinematics. These steps, which are indicated in green in Fig. 4, are carried out only at the beginning, prior to loading the model in VR. The object-mapping information can similarly be obtained from the CAD system via the CAD adapter. The model which has been processed in this way can then be loaded into the VR software. After the start of the interactive kinematics coupling (red arrows in Fig. 4), an automatically reduced, animated VR model with interactive kinematics is available.

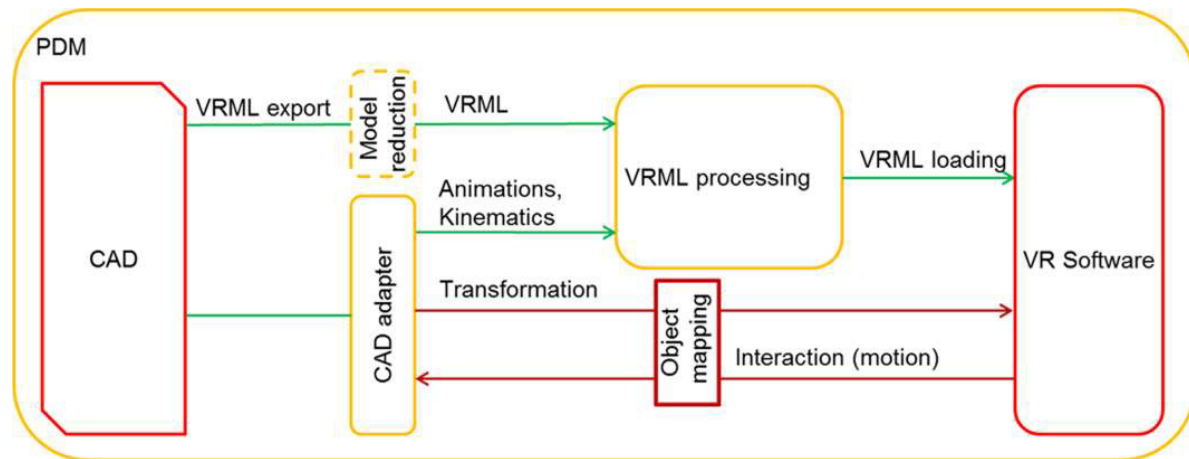


Fig. 4: Conversion workflow for an arbitrary CAD system

6. Validation

The presented method was validate in a proof of concept implementation using Siemens NX as a CAD system, Instant Reality as the VR system and Siemens Teamcenter as the PDM system. The whole conversion workflow was implanted within Siemens Teamcenter. Additional programs for handling certain parts of the conversion process were implemented and called in the background during the conversion process. Fig. 5 shows the interactive kinematic functionality realized by coupling Instant Realty with Siemens NX. In the shown example the user pulls at one rod causing the movement of the other rods and the cylinders that they are

connected through. The movement of the parts is calculated by the mechanism functionality of Siemens NX causing an update of the position of the parts in the VR model. The VR model moves correctly according to the defined kinematic definitions in the CAD model although no constraints or specific hierarchy was defined in the VR model.

7. Conclusion

The proposed concept presents a method which will in future permit a CAD model furnished with animations and kinematics to be used in the same way in a VR environment, without the necessity of complex manual model processing by an expert as it was previously the case. This significantly

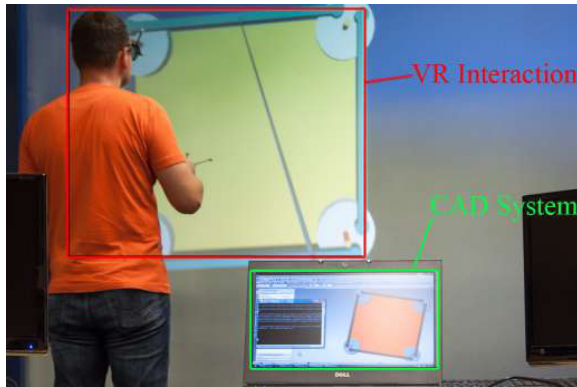


Fig. 5: Validation of interactive kinematic method

increases the chances that VR will increasingly become integral to product development in SMEs and that they might benefit from its potential to the same extent as it is currently the case in the automotive and aerospace industries. For the three significant points of model reduction, animation and kinematics transfer, there exists a proof of concept implementation. For the issues relating to model reduction and animation transfer, user-friendly modules based on currently available test results need to be developed. PDM implementation will manage all data produced during the conversion process and reference it to the original data while initiating and monitoring the conversion and processing of the 3D model and the start of interactive kinematics.

Using the method presented, VR-supported design reviews, for instance, can be carried out without the help of 3D/VR experts. Through integration into the PDM, VR support can be introduced seamlessly into the existing product development process as the design data can be processed and made usable for VR directly from the PDM. A further advantage is that the results of the VR-supported design review can also be directly held in the PDM and appropriate modification requests can be made. In view of the diversity of potential users with regard to the product development

process, the developed product, the CAD systems used, the PDM system installed and the existing IT network architecture, the method presented must be tailored to the company when it is installed.

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